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MATHEMATICAL CONTROL MODEL
Sensor fusion • Error • PID • VEL • Kinematics • Torque • Steering

1. Software Architecture

The control system is organized into sensor acquisition, sensor fusion, obstacle detection, PID motion control, speed control through VEL, motor control, and a safety override.

• **Sensor acquisition:** reads the four ultrasonic sensors.

• **Sensor fusion:** combines the 90° and 45° measurements into the control error.

• **Obstacle detection:** uses the front ultrasonic sensor for the safety condition.

• **PID motion control:** converts ERROR into a steering correction.

• **VEL control:** defines the numerical speed command used by the program.

• **Motor control:** applies the resulting command to the drive motors.

• **Safety override:** takes priority when the front obstacle threshold is reached.

2. Software Execution Flow

→ Initialize the drive system.

→ Read the front ultrasonic sensor.

→ Store the measurement as **ARD**.

→ If **ARD < 25 cm**, stop, wait a few milliseconds, and reverse **0.5 motor rotations**.

→ Otherwise, read the four lateral ultrasonic sensors.

→ Calculate the sensor differences and then ERROR.

→ Set the programmed VEL numerical command.

→ Calculate/update the PID terms.

→ Apply the motor command.

3. Ultrasonic Sensor Configuration

The robot uses exactly four lateral ultrasonic sensors: **two sensors at 90°** and **two sensors at 45°**. The 90° pair provides direct lateral measurements, while the 45° pair provides diagonal measurements.

$$S_{90,1}, S_{90,2} \rightarrow \textbf{direct lateral distances}$$
$$S_{45,1}, S_{45,2} \rightarrow \textbf{diagonal distances}$$

4. Error Calculation

4.1 Sensor differences

$$E_{90} = S_{90,1} - S_{90,2}$$

This difference represents the imbalance measured by the two 90° sensors.

$$E_{45} = S_{45,1} - S_{45,2}$$

This difference represents the imbalance measured by the two 45° sensors.

4.2 Final control error

$$ERROR = (S_{90\_1} - S_{90\_2}) - (S_{45\_1} - S_{45\_2}) + 100$$

$$ERROR = E_{90} - E_{45} + 100$$

The **+100** is a calibration/control offset established for the sensor configuration. It shifts the numerical error to the reference value used by the robot's control system. It is a control-system offset, not a physical distance, force, or torque.

5. PID Motion Controller

$$I(t) = I(t-1) + ERROR(t)$$

Integral: accumulates error over time, allowing correction of persistent deviation.

$$D(t) = ERROR(t) - ERROR(t-1)$$

Derivative: measures how quickly the error changes.

$$PID\_Output = K_p \cdot ERROR + K_i \cdot I + K_d \cdot D$$

- **P:** responds to the current error.
- **I:** responds to accumulated error.
- **D:** responds to the change in error.
- The gains Kp, Ki and Kd are tuned through real-track testing.

6. Speed Control — VEL

VEL is treated as a **numerical value** in the control program. The programmed command uses the 40–50 numerical offset:

$$VEL\_command = VEL\_base + 40$$

$$VEL\_command = VEL\_base + 50$$

The selected programmed value is therefore **VEL_base + 40** or **VEL_base + 50**, according to the corresponding robot condition/configuration.

7. Front Ultrasonic Safety System

$$\text{If } ARD < 25 \text{ cm} \rightarrow STOP \rightarrow WAIT \rightarrow REVERSE\ 0.5 \text{ rotations}$$

The front-obstacle condition has priority over the normal steering loop. The reverse command is fixed at **0.5 motor rotations**.

8. Motor Kinematics & Linear Displacement

8.1 Wheel circumference

$$C = \pi D$$

For D = 56.0 mm:

$$C = \pi(56.0) \approx 175.93 \text{ mm}$$

8.2 Distance from motor angle

$$d = (\theta / 360°) \cdot C$$

$$d_{1°} = 175.93 / 360 \approx 0.4887 \text{ mm/°}$$

$$\theta = d / 0.4887$$

These equations connect motor rotation angle with ideal linear wheel travel.

9. Reverse Maneuver Analysis

$$0.5 \text{ rotations} = 180°$$

$$d = (180 / 360) \cdot 175.93 \approx 87.97 \text{ mm}$$

Ideal wheel-based displacement is approximately **87.97 mm**. Actual displacement can differ because of tire slip, friction, deformation, drivetrain losses, and surface conditions.

10. Motor Torque

The robot uses **LEGO EV3 Medium Motors**. Reference running torque: **8 N·cm = 0.08 N·m**. Stall torque: **12 N·cm = 0.12 N·m**; stall torque represents a blocked-rotor condition rather than normal driving.

$$\tau = F \cdot r$$

Wheel radius: $r = 56/2 = 28$ mm $= 0.028$ m.

$$F = \tau / r = 0.08 / 0.028 \approx 2.86 \text{ N}$$

This is an ideal tangential-force estimate. Actual tractive force depends on drivetrain efficiency, gearing, friction, load, and wheel slip.

11. Steering Geometry

Wheelbase **L = 24.0 cm**; track width **W = 18.0 cm**. The geometry is consistent with an Ackermann-type turning model.

$$\delta_{internal} = \operatorname{atan}\left(L / ((L / \tan(\alpha)) - W/2) \right)$$

$$\delta_{external} = \operatorname{atan}\left(L / ((L / \tan(\alpha)) + W/2) \right)$$

The inner and outer wheels follow different turning radii.

12. Diagonal Ultrasonic Geometry

$$Y_{wall} = D_{diagonal} \cdot \sin(45°) \approx 0.7071 D_{diagonal}$$

$$X_{forward} = D_{diagonal} \cdot \cos(45°) \approx 0.7071 D_{diagonal}$$

The diagonal measurement is decomposed into lateral and forward components.

13. Engineering Decisions

- Use four ultrasonic sensors with two 90° and two 45° orientations.
- Use sensor differences to create the control error.
- Use the +100 calibration/control offset in the programmed ERROR equation.
- Use PID to transform ERROR into steering correction.
- Keep VEL as a numerical command with the programmed +40 / +50 offset.
- Give the front-obstacle safety routine priority.
- Base kinematic calculations on the robot's actual wheel dimensions.
- Avoid unsupported numerical assumptions.

14. Testing & Validation

The control system is validated through iterative testing on the real track. Sensor behavior, ERROR response, PID gains, VEL values, obstacle response, and steering behavior are evaluated and adjusted from observed robot performance.

15. Future Improvements

- Add stronger sensor filtering while preserving the existing control equation.
- Evaluate adaptive PID parameters for different track sections.
- Optimize programmed VEL values for stability and speed.
- Add data logging for ERROR, PID output, VEL, and sensor readings.
- Measure drivetrain torque and efficiency experimentally.
- Model slip and friction compensation.
- Compare parameter sets using repeatable track tests.

16. Reference Values

Parameter	Value	Meaning
Wheelbase L	24.0 cm	Distance between front/rear axle lines
Track width W	18.0 cm	Lateral distance between wheel paths
Wheel diameter D	56.0 mm	Drive-wheel diameter
Wheel circumference C	175.93 mm	πD
Motor resolution	1°	EV3 Medium Motor rotation sensor
Linear distance / degree	0.4887 mm/°	175.93 / 360
Front critical distance	25 cm	Safety threshold: ARD < 25 cm
Reverse command	0.5 rotations	Ideal displacement ≈ 87.97 mm
Running torque	0.08 N·m	Reference operating torque
Stall torque	0.12 N·m	Blocked-rotor reference
Ideal wheel tangential force	≈ 2.86 N	0.08 / 0.028
ERROR offset	+100	Calibration/control offset
VEL command	VEL_base + 40 / +50	Programmed numerical speed command